

Indian Institute of Technology Jodhpur
MAL1020
Tutorial Sheet 7

1. Find the general solution of the following differential equations:

(a) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 8y = 0$.

(b) $\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 8y = 0$.

(c) $\frac{d^4y}{dx^4} - 5\frac{d^3y}{dx^3} + 6\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 8y = 0$.

2. For certain values of the constant m , the function f defined by $f(x) = e^{mx}$ is a solution of the differential equation

$$\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 12y = 0.$$

Determine all such values of m .

3. Solve the following non homogeneous linear differential equations with constant coefficients (without method of undetermined coefficient) :

(a) $y'' - 3y' + 2y = e^{5x}$

(b) $y'' - y = 2 + 5x$

(c) $y'' - 2y' + 5y = e^{-x}$

(d) $y''' + 3y'' + 3y' + y = e^{-x}$

4. Show that the first-order differential equation

$$\left| \frac{dy}{dx} \right| + |y| + 1 = 0$$

has no real solutions.

5. Show that the first-order differential equation

$$\left(\frac{dy}{dx} \right)^2 - 4y = 0$$

has a one-parameter family of solutions of the form $f(x) = (x + c)^2$, where c is a constant.

6. (Method of undetermined coefficients) Solve the following differential equations:

(a) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^{4x}$.

(b) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = x^2e^x$.

(c) $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 8y = x^3 + x + e^{-2x}$.

7. Solve following problems using method of undetermined coefficients:

(a) $y'' - 4y' - 12y = 3e^{5x}$

(b) $y'' - 4y' - 12y = \sin(2x)$

(c) $y'' - 4y' - 12y = 2x^3 - x + 3$

(d) $y'' - 4y' - 12y = xe^{4x}$

(e) $y'' - 4y' - 12y = e^{6x}$

8. Figure out the form of particular solution of the given differential equations using method of undetermined coefficients:

(a) $y'' + 3y' - 28y = 7x + e^{-7x} - 1$

(b) $y'' - 100y = 9x^2e^{10x} + \cos(x) - x\sin(x)$

(c) $4y'' + y = e^{-2x}\sin(\frac{x}{2}) + 6x\cos(\frac{x}{2})$

(d) $4y'' + 16y' + 17y = e^{-2x}\sin(\frac{x}{2}) + 6x\cos(\frac{x}{2})$

(e) $y'' + 8y' + 16y = e^{-4x} + (x^2 + 5)e^{-4x}$

9. Solve following first order differential equations:

(a) $\left(\frac{dy}{dx}\right)^2 - 7\frac{dy}{dx} + 12 = 0$

(b) $\left(\frac{dy}{dx}\right)^3 + 2x\left(\frac{dy}{dx}\right)^2 - y^2\left(\frac{dy}{dx}\right)^2 - 2xy\frac{dy}{dx} = 0$

(c) $\left(\frac{dy}{dx}\right)^2 + 2y\cot(x)\frac{dy}{dx} = y^2$

(d) $\left(\frac{dy}{dx}\right)^2 - ax^3 = 0$

10. Solve the following differential equations:

(a) Show that a solution of equation $y = x\left(\frac{dy}{dx}\right) + f\left(\frac{dy}{dx}\right)$ is of form $y = cx + f(c)$ for some c .

(b) $\frac{dy}{dx} = \log\left(x\left(\frac{dy}{dx}\right) - y\right)$

(c) $\cos(y)\sin\left(x\left(\frac{dy}{dx}\right)\right) = \sin(y)\cos\left(x\left(\frac{dy}{dx}\right)\right) + \frac{dy}{dx}$

(d) $y = \left(1 + \left(\frac{dy}{dx}\right)\right)x + \left(\frac{dy}{dx}\right)^2$

(e) $xy^2\left(2 + \left(\frac{dy}{dx}\right)^2\right) = 2y^3\frac{dy}{dx} + x^3$